



The Islamic University of Gaza
Faculty of Engineering
Department of Computer Engineering



ECOM 1101 - Digital Design

Lab #1 Logic Gates

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Introduction

What is Logic Design?

Logic design is a fundamental discipline within digital electronics that focuses on the representation, analysis, and implementation of systems based on binary values (0 and 1). It involves the use of Boolean algebra and logic gates (such as AND, OR, and NOT) to design circuits capable of performing specific computational or decision-making tasks. Through logic design, complex digital systems—ranging from simple combinational circuits to advanced processors—can be systematically constructed and optimized for efficiency, reliability, and scalability.

ON/OFF (0 &1) Concept

In digital systems, information is represented using the binary number system, which consists of only two possible values: 0 and 1. These values, known as binary digits or bits, correspond to two distinct physical states within electronic circuits, such as low and high voltage levels. Typically, 0 represents the OFF or false state, while 1 represents the ON or true state. This simple yet powerful representation forms the foundation of all digital logic, enabling computers and electronic systems to perform calculations, store data, and make logical decisions.



What is Logic gate ?

Logic gates are essential components in digital electronics, responsible for processing and manipulating binary information. They are electronic circuits that perform logical operations on binary inputs, producing binary outputs based on defined rules. These gates include fundamental types such as AND, OR, NOT, XOR, NAND, and NOR gates, each with its unique behavior and truth table. By combining these gates, complex operations can be performed, enabling the design and implementation of various digital systems. Logic gates are the backbone of modern computing, playing a crucial role in everything from microprocessors and memory units to communication systems and control circuits. Understanding logic gates is fundamental to comprehend the functioning of digital systems and advanced technologies.

What is the Truth Table?

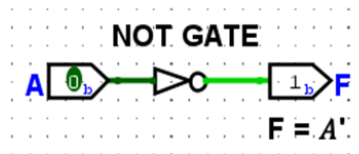
A truth table is a tabular representation that shows all possible input combinations and their corresponding output values for a logical expression or gate. It provides a concise summary of how the expression or gate behaves under different input scenarios. Each row in the table represents a unique input combination, while the columns represent the input variables and the output(s)

Basic Logic Gates

NOT GATE:

A NOT gate, also known as an inverter, is a basic logic gate that has one input and one output. It produces the complement (opposite) of the input signal. The NOT gate is represented by the symbol "¬" or "NOT."

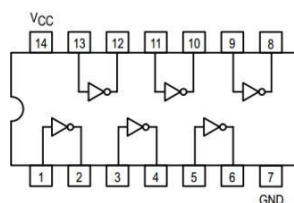
- ☐ Boolean expression :bar
($F = A'$).
- ☐ Logic Diagram



- ☐ IC Number: **7404**
- ☐ Truth Table:
When the input is "0," the output is "1"
When the input is "1," the output is "0"

Input	Output
0	1
1	0

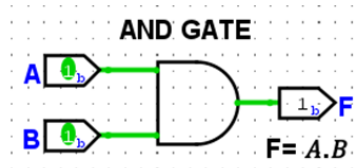
- ☐ Pin Diagram



AND GATE:

An AND gate is a fundamental logic gate with two inputs or more and a single output. The AND gate is represented by the symbol "." or "AND."

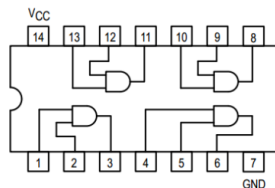
- ☐ Boolean expression :dot
($F = A.B = (AB)$)
- ☐ Logic Diagram



- ☐ IC Number: **7408**
- ☐ Truth Table:
An AND gate outputs "**1**" only when **all inputs are "1"**
If any of the inputs are "0" the output is "0"

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

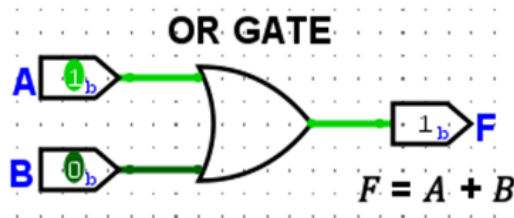
- ☐ Pin Diagram



OR GATE:

An OR gate is a basic logic gate with two inputs or more and a single output. The OR gate is represented by the symbol "+" or "OR."

- ☐ Boolean expression : plus sign
($F = A + B$).
- ☐ Logic Diagram



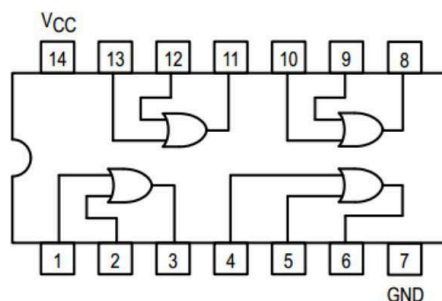
- ☐ IC Number: 7432

- ☐ Truth Table:

The **OR** gate outputs **1** when **at least one of its inputs is 1** (including when all inputs are 1).
If **all inputs are 0**, the output is **0**

A	B	F
0	0	0
0	1	1
1	0	1
1	1	1

- ☐ Pin Diagram



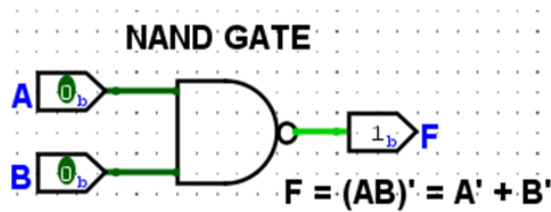
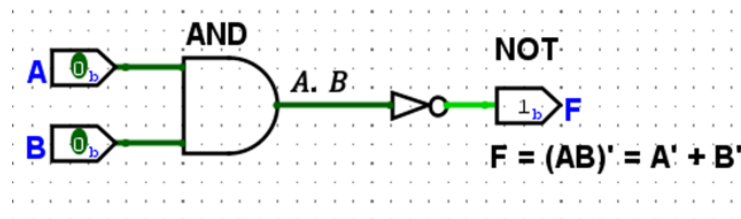
NAND GATE

A NAND gate is a fundamental logic gate with two inputs or more and a single output.

□ Boolean expression :($F = (AB)' = A' + B'$).

□ Logic Diagram

It is essentially an AND gate followed by a NOT gate.



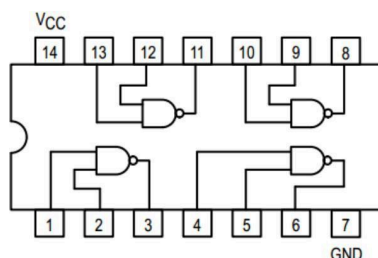
□ IC Number: 7400

□ Truth Table:

A **NAND gate** outputs "**0**" **only** when **all inputs are "1"**
If **any input is "0"** the output is "**1**"

A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

□ Pin Diagram

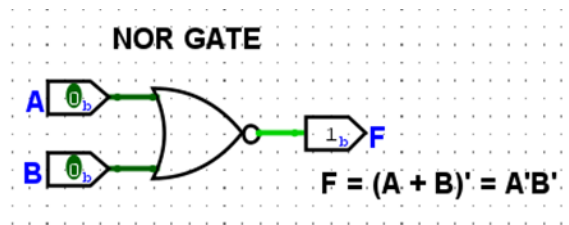
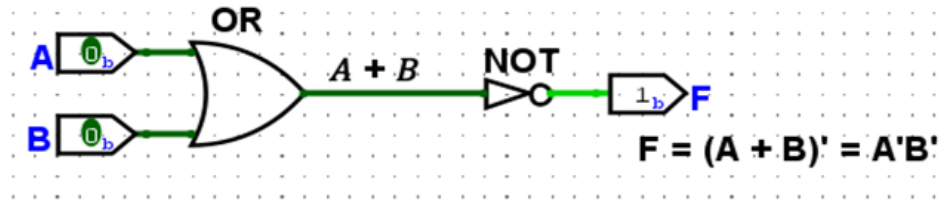


NOR GATE

A NOR gate is a fundamental logic gate with two inputs or more and a single output.

- Boolean expression : $(F = (A + B)' = (A'B'))$
- Logic Diagram

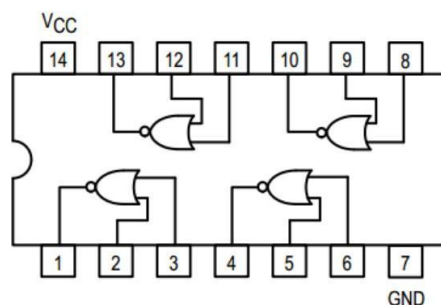
It is essentially an OR gate followed by a NOT gate.



- IC Number: 7402
- Truth Table:
NOR gate produces a "1" output only when both inputs are "0"
Otherwise; the output is "0"

A	B	F
0	0	1
0	1	0
1	0	0
1	1	0

- Pin Diagram

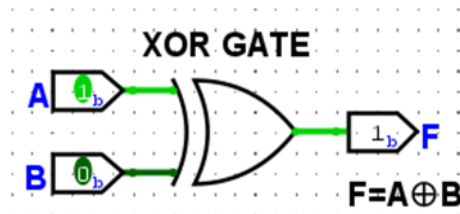


XOR GATE:

An XOR gate is a logic gate that has two or more inputs and a single output .
It is commonly used for bitwise operations and data manipulation

- Boolean expression :
($F = A \oplus B = A'B + AB'$).

- Logic Diagram



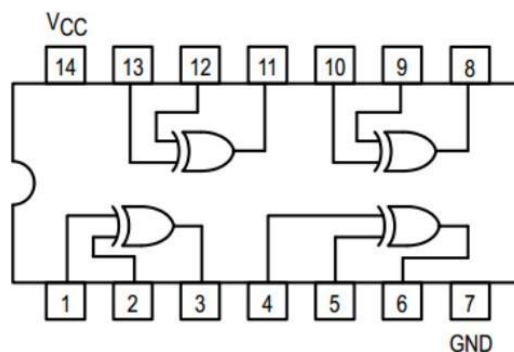
- IC Number: **7486**

- Truth Table:

The XOR gate produces an output of 1 when the number of inputs set to 1 is odd,
and produces an output of 0 when the number of inputs set to 1 is even.
In other words, the output is 1 when the inputs are different, and the output is 0
when the inputs are the same

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

- Pin Diagram



XNOR GATE:

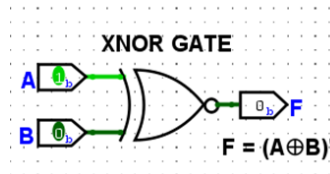
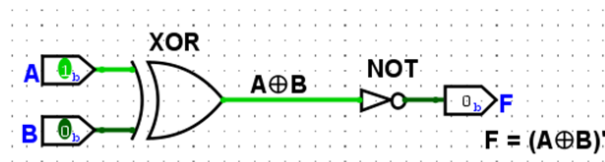
An XNOR gate is a logic gate that has two or more inputs and a single output

- Boolean expression :

$$(F = (A \oplus B)' = (A + B') (A' + B)).$$

- Logic Diagram

It is essentially an XOR gate followed by a NOT gate.



- IC Number: **747266**

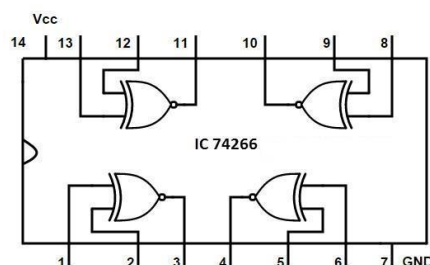
- Truth Table

The XNOR gate produces an output of 0 when the number of inputs set to 1 is odd, and produces an output of 1 when the number of inputs set to 1 is even.

In other words, the output is 0 when the inputs are different, and the output is 1 when the inputs are the same.

A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

- Pin Diagram



Boolean Expressions

Boolean expressions are a fundamental aspect of logic and programming. They combine variables, logical operators, and constants to produce logical results. A Boolean expression can be evaluated to either true or false based on the values assigned to the variables. Logical operators such as AND, OR, and NOT are used to manipulate variables within the expression. By applying the laws of Boolean algebra and simplifying expressions, programmers can create logical and efficient algorithms.

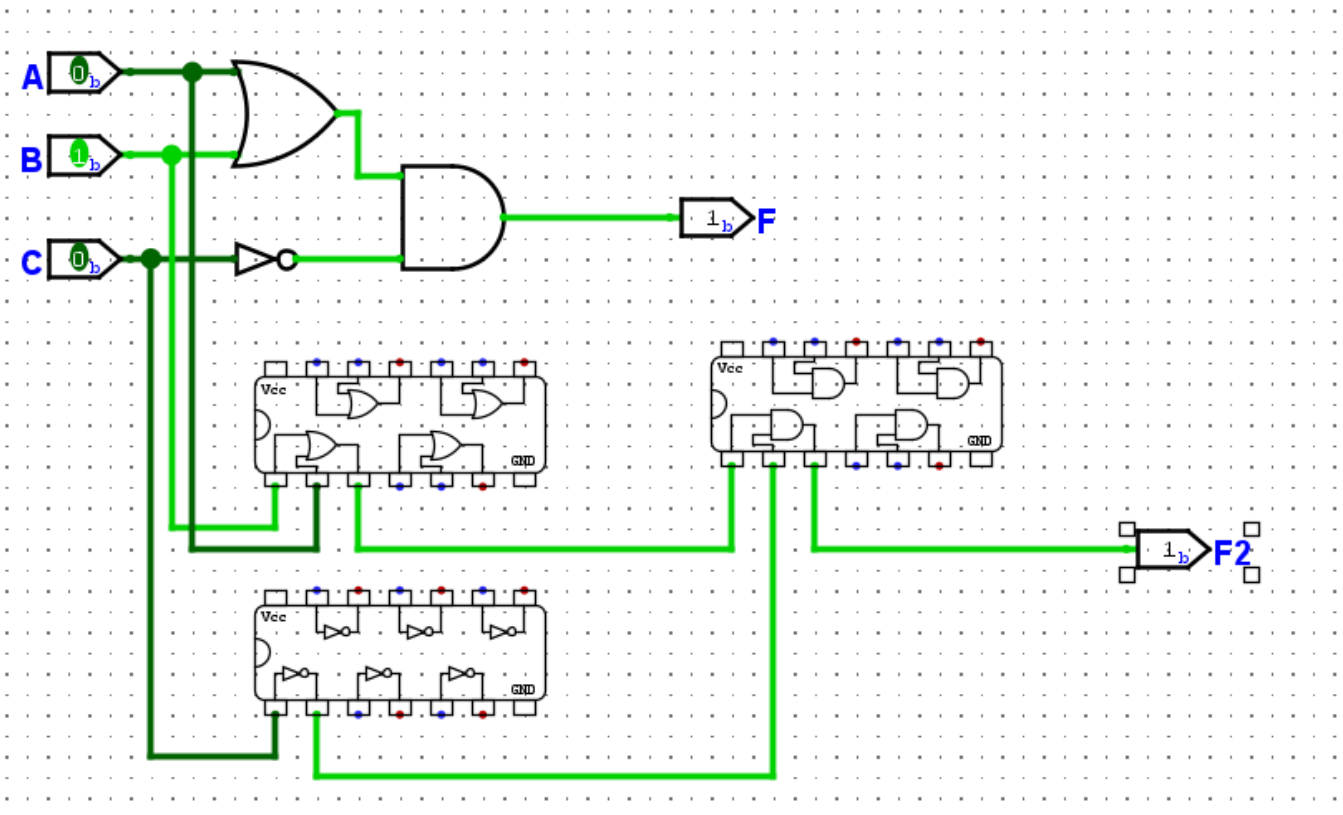
Operations Order

The order of operations for logic gates, also known as logic operations, refers to the sequence in which different logic gates are evaluated within a logical expression. The specific order depends on the precedence of logic operations and the logical operators involved. and here is the order of the operations in Boolean expressions (Parentheses > NOT > AND > OR).

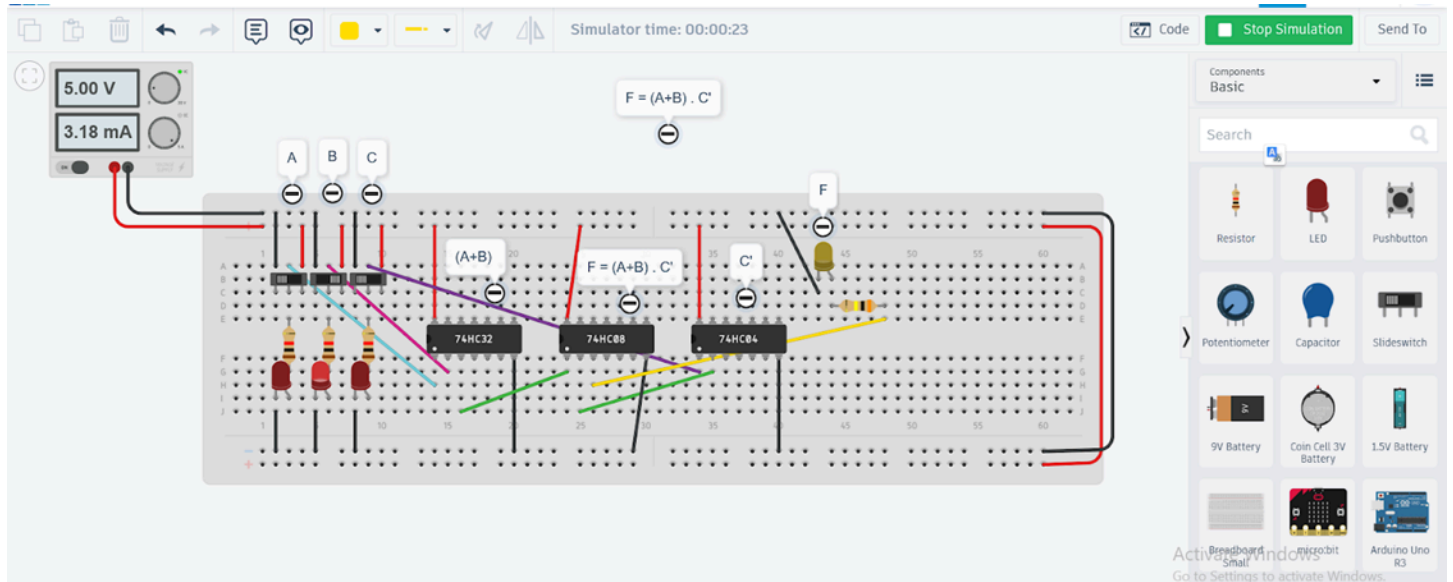
Example of Boolean Expressions

EX#1: $F = (A+B) \cdot C'$

Logisim:



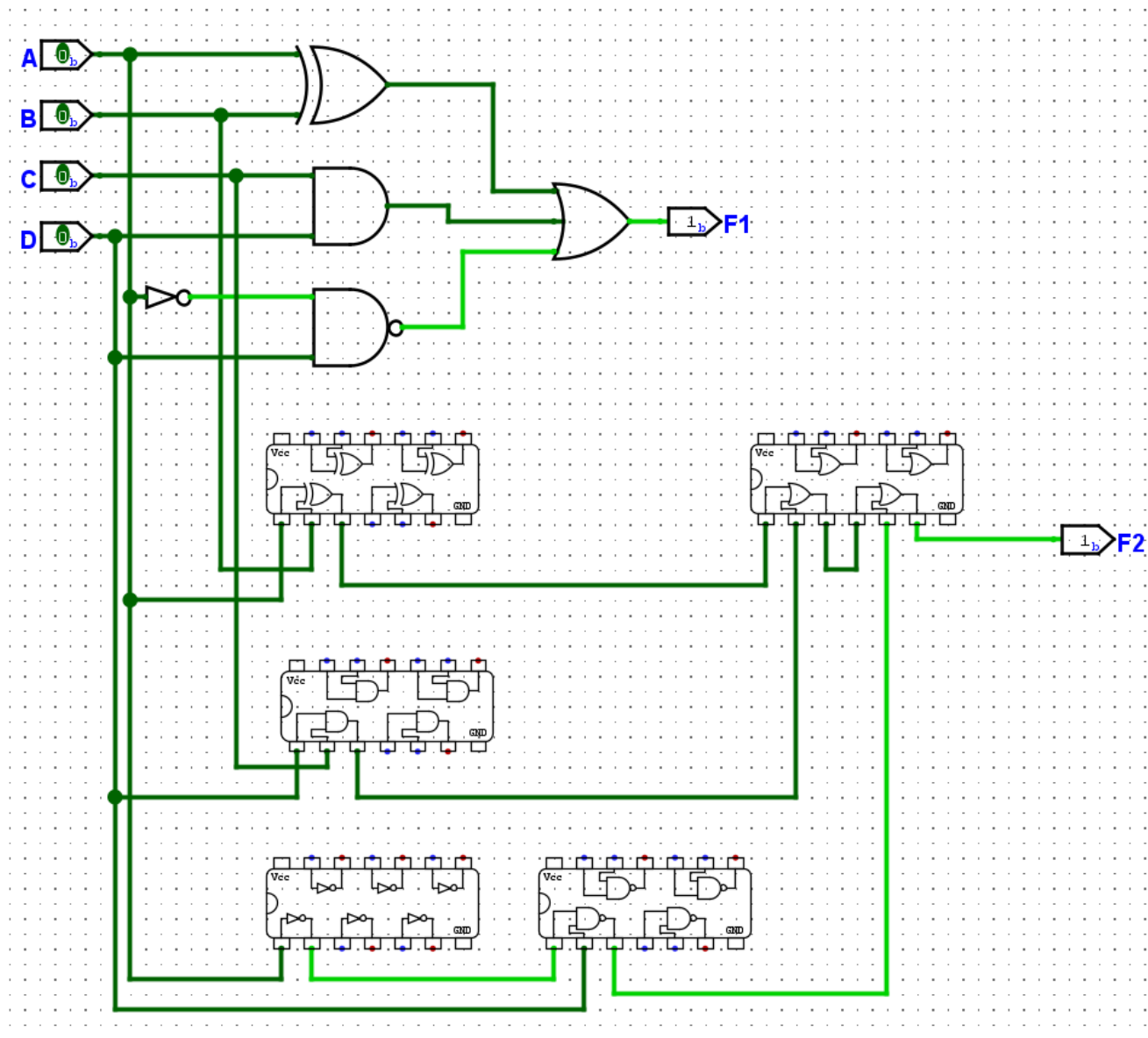
Tinkercad:



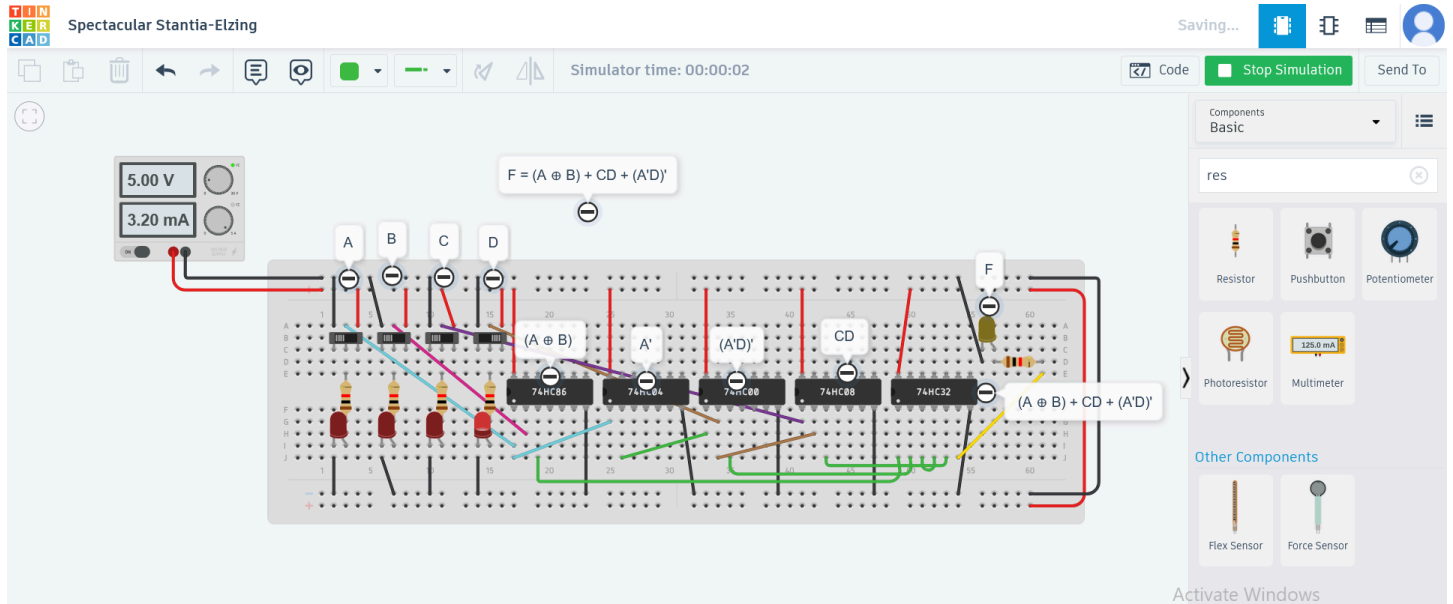
EX#2

$$F = (A \oplus B) + CD + (A'D)'$$

Logisim:



Tinkercad:



Homework:

Implement the following boolean expression:

1. $F = (XY'Z + (Y \oplus W)')'$
2. $F = (A+B)(C+D)+A'$